

Homework 4

Danen M. Pruitt

ENS, AFIT

OPER 646: Reinforcement Learning

May 2026

Contents

1	Introduction	2
2	SARSA	2
3	Q-Learning	3
4	n-step SARSA	5
5	Algorithm Comparison	7
6	Momentum Policy Comparison	7
7	Reflection	9
8	AI Usage	10
9	Appendix: Reinforcement Learning Hyperparameter Design Runs & Statistical Reports	10

1 Introduction

In this assignment we will be considering the Gymnasium environment *MountainCar-v0* and solving the agent's sequential decision problem using SARSA, Q-learning, and n-step SARSA. Across all of the algorithm implementations we configured them with the following stepsize TD learning rates.

$$\alpha_n = \alpha_a \left(\frac{1}{(n+1)^{\alpha_b}} \right)$$

$$\epsilon_n = \epsilon_a \left(\frac{1}{(n+1)^{\epsilon_b}} \right)$$

We utilized a Latin hypercube design to tune the hyperparameters $\alpha_a, \alpha_b, \epsilon_a, \epsilon_b$, and Q^{init} . Each was tuned with 60 design runs at 10 replications with 1000 episodes per replication. In each of the sections below, tables and figures will be presented. For brevity we will only look at the OLS and ANOVA results for each hyperparameter individually. Interaction terms can be found in the appendix.

2 SARSA

Table 1: Top 10 Results for SARSA.onpolicy

Run Index	α_a	α_b	ϵ_a	ϵ_b	qinit	Sup EETDR	Sup EETDR hw	Mean Max EETDR	Mean Max EETDR ME	Time-Avg EETDR	Time-Avg EETDR ME	Secs per run	Alg Score
15	0.242	0.125	0.375	0.358	0.142	-120.150	6.392	-123.840	5.794	-168.541	4.702	233.409	-147.078
14	0.592	0.175	0.358	0.275	0.942	-114.700	7.026	-121.720	6.493	-173.997	4.241	239.248	-148.223
11	0.558	0.342	0.858	0.308	0.308	-120.900	6.641	-126.610	7.050	-172.213	3.284	237.256	-150.395
41	0.992	0.358	0.175	0.325	0.542	-125.950	6.997	-131.360	8.752	-161.449	4.939	222.038	-150.622
43	0.825	0.508	0.042	0.758	0.492	-130.300	13.073	-128.290	6.270	-172.981	4.064	233.549	-151.776
6	0.275	0.108	0.742	0.425	0.175	-127.650	5.670	-126.750	7.231	-174.418	4.187	241.776	-151.831
39	0.842	0.542	0.625	0.908	0.225	-115.050	6.835	-125.830	7.910	-177.920	4.742	238.083	-153.309
21	0.342	0.142	0.125	0.825	0.442	-127.100	6.089	-136.030	9.456	-160.759	4.919	221.156	-153.563
50	0.092	0.008	0.008	0.342	0.325	-127.800	5.331	-129.960	7.608	-171.060	8.509	233.948	-154.368
42	0.442	0.392	0.025	0.442	0.608	-115.850	7.017	-130.890	8.335	-173.129	4.163	234.303	-154.452

Above is a table of the top 10 design runs for the SARSA on-policy algorithm with respect to algorithm score with the best design run's superlative policy EETDR of -124.25. The design run also gave us the following learning curve.

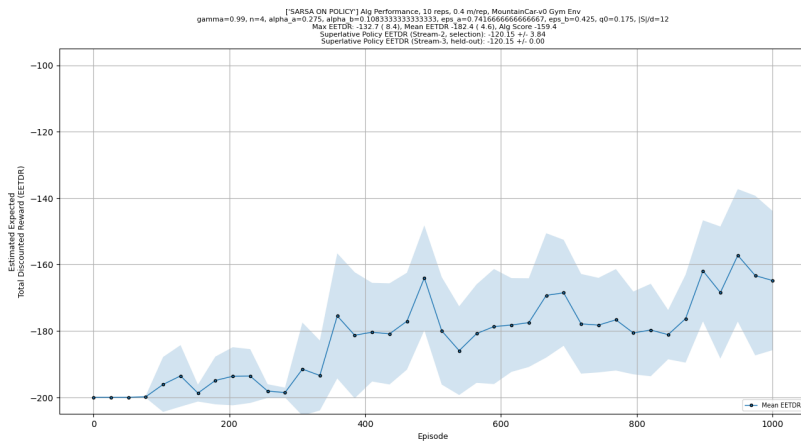


Figure 1: Learning Curve for Best SARSA run

This figure illustrates the on-policy nature of the SARSA algorithm in the wide variance shaded regions. Since our epsilon values remain relatively large the on-policy algorithm explores more making the variance large. Below is a table of the OLS and ANOVA results for the SARSA algorithm.

OLS and ANOVA Regression Results

(a) OLS Model Fit Summary		(b) ANOVA Type II Effects Table				
Metric	Value	Variable	sum_sq	df	F	PR(> F)
R^2	0.767	α_a	91.614	1.000	0.727	0.399
Adj. R^2	0.647	α_b	104.986	1.000	0.834	0.367
F-statistic	6.415	ϵ_a	86.722	1.000	0.689	0.412
Prob (F-stat)	4.09×10^{-07}	ϵ_b	1018.373	1.000	8.087	0.007
Log-Likelihood	-217.290	qinit	67.197	1.000	0.534	0.469
AIC	476.600					
BIC	520.600					

(c) OLS Coefficient Estimates						
Variable	coef	std err	t	P > t	[0.025	0.975]
Intercept	-189.270	14.027	-13.494	0.000	-217.642	-160.899
α_a	28.662	33.604	0.853	0.399	-39.310	96.633
α_b	-26.545	29.073	-0.913	0.367	-85.351	32.261
ϵ_a	24.784	29.866	0.830	0.412	-35.626	85.194
ϵ_b	80.698	28.378	2.844	0.007	23.298	138.098
qinit	23.925	32.753	0.730	0.469	-42.324	90.174

From this table we see the only statistically significant variable is ϵ_b . We are confident that it is the most significant as our model has high R^2 and Adj R^2 values telling us that it is a well fit and explanatory model. We shouldn't discount the other variables but it is not surprising that the algorithm, which is a TD algorithm, is more sensitive to the epsilon decay rate than learning rates or initial values. The higher our rate of decay the more we exploit rather than explore the policy we are currently following.

3 Q-Learning

Table 3: Top 10 Results for Q-learning_offpolicy

Run Index	α_a	α_b	ϵ_a	ϵ_b	qinit	Sup EETDR	Sup EETDR hw	Mean Max EETDR	Mean Max EETDR ME	Time-Avg EETDR	Time-Avg EETDR ME	Secs per run	Alg Score
43	0.825	0.508	0.042	0.758	0.492	-126.050	6.029	-126.880	5.605	-168.781	4.183	250.749	-148.676
15	0.242	0.125	0.375	0.358	0.142	-127.950	6.586	-123.900	4.504	-174.079	5.080	263.673	-148.706
14	0.592	0.175	0.358	0.275	0.942	-118.800	7.693	-124.490	6.095	-176.660	4.090	269.760	-150.651
53	0.692	0.525	0.575	0.875	0.458	-118.600	6.044	-126.100	6.346	-174.841	3.157	258.703	-150.667
11	0.558	0.342	0.858	0.308	0.308	-125.150	8.679	-127.610	8.246	-170.753	4.204	262.617	-151.497
41	0.992	0.358	0.175	0.325	0.542	-127.550	5.287	-137.330	4.676	-160.762	5.582	242.912	-151.742
27	0.658	0.442	0.925	0.375	0.375	-123.300	8.339	-132.000	6.896	-170.493	2.638	258.634	-152.590
36	0.192	0.242	0.225	0.608	0.192	-119.400	8.021	-130.350	7.251	-171.768	3.982	256.396	-152.861
42	0.442	0.392	0.025	0.442	0.608	-119.400	5.838	-130.760	7.386	-171.793	3.640	252.681	-153.061
32	0.075	0.042	0.308	0.175	0.475	-118.950	7.355	-129.340	6.707	-175.987	2.930	266.639	-153.195

Above is a table of the top 10 design runs for the Q-Learning off-policy algorithm with respect to algorithm score with the best design run's superlative policy EETDR of -126.05. The design run also gave us the following learning curve.

$\epsilon_a=0.99$, $n=4$, $\alpha_a=0.6583333333333333$, $\alpha_b=0.4416666666666666$, $\epsilonps_a=0.925$, $\epsilonps_b=0.375$, $q0=0.37$
 Max EETDR: -134.5 (7.2), Mean EETDR -179.0 (4.2), Alg Score -158.3
 Superlative Policy EETDR (Stream-2, selection): -126.05 +/- 4.11
 Superlative Policy EETDR (Stream-3, held-out): -126.05 +/- 0.00

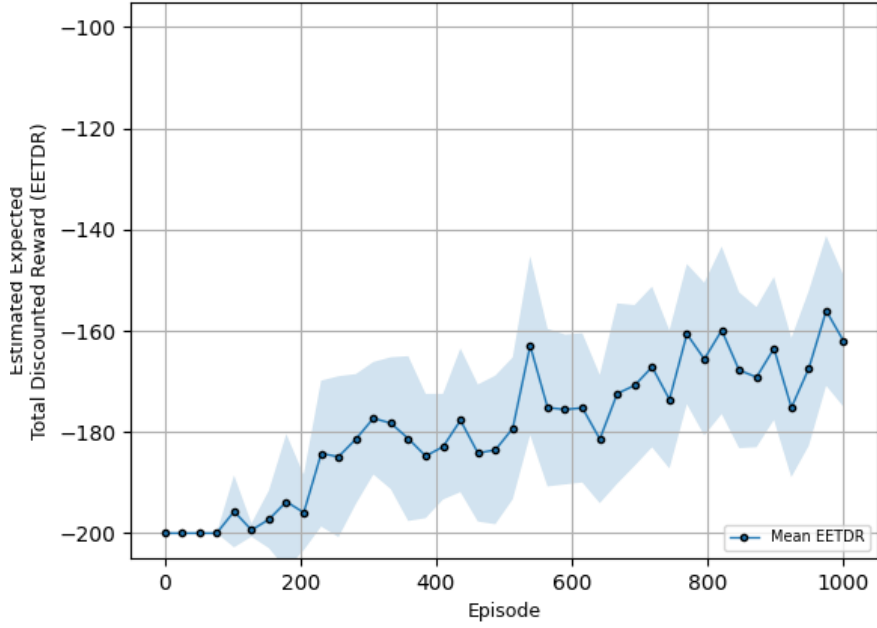


Figure 2: Learning Curve for Best Q-Learning run

Unlike the earlier SARSA algorithm, the Q-learning algorithm starts to learn the optimal policy earlier showing the power of the off-policy algorithm early on. The oscillations in the middle of the figure are likely due to the large exploration rate of ϵ_a forcing the agent to take more random actions rather than greedy ones but there is still steady improvement. Finally we notice that the variance shading is tighter due to the disconnect between the behavior policy and the target policy showing its ability to identify and maintain a high performing greedy policy even with the environmental noise.

In the tables below we see that again the only statistically significant variable is ϵ_b . It takes the highest coefficient value as well as the highest sum of squares value. Again, Q-learning is a TD learning algorithm so the rate at which we move from exploration to exploitation is extremely important to how well the algorithm performs and progresses in learning.

OLS and ANOVA Regression Results

(a) Q-Learning Model Fit Summary

Metric	Value
R^2	0.786
Adj. R^2	0.676
F -statistic	7.168
Prob (F -stat)	9.38×10^{-08}
Log-Likelihood	-214.630
AIC	471.300
BIC	515.200

(b) Q-Learning ANOVA Type II Effects Table

Variable	sum_sq	df	F	PR(> F)
α_a	219.424	1.000	1.904	0.176
α_b	44.162	1.000	0.383	0.540
ϵ_a	74.924	1.000	0.650	0.425
ϵ_b	1311.899	1.000	11.382	0.002
qinit	107.902	1.000	0.936	0.339

(c) Q-Learning Coefficient Estimates

Variable	coef	std err	t	$P > t $	[0.025	0.975]
Intercept	-196.515	13.419	-14.645	0.000	-223.657	-169.372
α_a	44.357	32.149	1.380	0.176	-20.669	109.384
α_b	-17.216	27.814	-0.619	0.540	-73.475	39.042
ϵ_a	23.036	28.572	0.806	0.425	-34.756	80.829
ϵ_b	91.592	27.149	3.374	0.002	36.679	146.506
qinit	30.318	31.334	0.968	0.339	-33.062	93.697

4 n-step SARSA

Below is a table of the top 10 design runs for the n-step SARSA algorithm with the additional column of n-steps. Here the best design run scored the highest of all three with a superlative policy EETDR of -115.45. The learning curve for this design run is also shown below. In comparison with the TD(0)

Table 5: Top 10 Results for n_step_SARSA_results.csv

Run Index	α_a	α_b	ϵ_a	ϵ_b	qinit	n_steps	Sup EETDR	Sup EETDR hw	Mean Max EETDR	Mean Max EETDR ME	Time-Avg EETDR	Time-Avg EETDR ME	Secs per run	Alg Score
11	0.558	0.342	0.858	0.308	0.308	5	-115.450	0	-122.730	7.453	-145.831	5.649	200.855	-127.732
20	0.875	0.325	0.142	0.392	0.958	8	-119.550	0	-131.060	9.295	-140.680	9.016	184.465	-131.608
41	0.992	0.358	0.175	0.325	0.542	3	-118.100	0	-130.990	8.047	-147.120	5.349	187.639	-131.848
27	0.658	0.442	0.925	0.375	0.375	7	-119.000	0	-138.250	6.894	-146.375	6.674	193.529	-132.620
1	0.425	0.625	0.458	0.625	0.158	9	-116.350	0	-139.030	12.529	-151.154	7.306	194.182	-133.194
31	0.625	0.558	0.158	0.658	0.808	3	-117.550	0	-121.210	3.040	-153.135	3.660	193.853	-133.248
15	0.242	0.125	0.375	0.358	0.142	6	-118.550	0	-137.990	7.958	-150.900	4.874	198.906	-133.440
30	0.925	0.792	0.842	0.508	0.725	6	-116.950	0	-124.290	7.358	-155.865	3.670	198.146	-133.984
6	0.275	0.108	0.742	0.425	0.175	4	-116.350	0	-129.550	6.894	-156.351	4.377	202.107	-134.101
36	0.192	0.242	0.225	0.608	0.192	2	-117.200	0	-131.560	8.738	-153.868	7.269	192.786	-134.775

algorithm of SARSA, n-step SARSA also takes an early rise in EETDR with a narrower variance shading throughout the 1000 episodes. The higher variance of the n-step method is shown in the large spikes and drops during the middle episodes and then plateaus towards the end with the tight variance shading. This graph shows us the benefits of the low-bias, high-variance nature of the algorithm. Due to the n-steps used to learn about the environment, the algorithm has better feedback as to what is working and what is not.

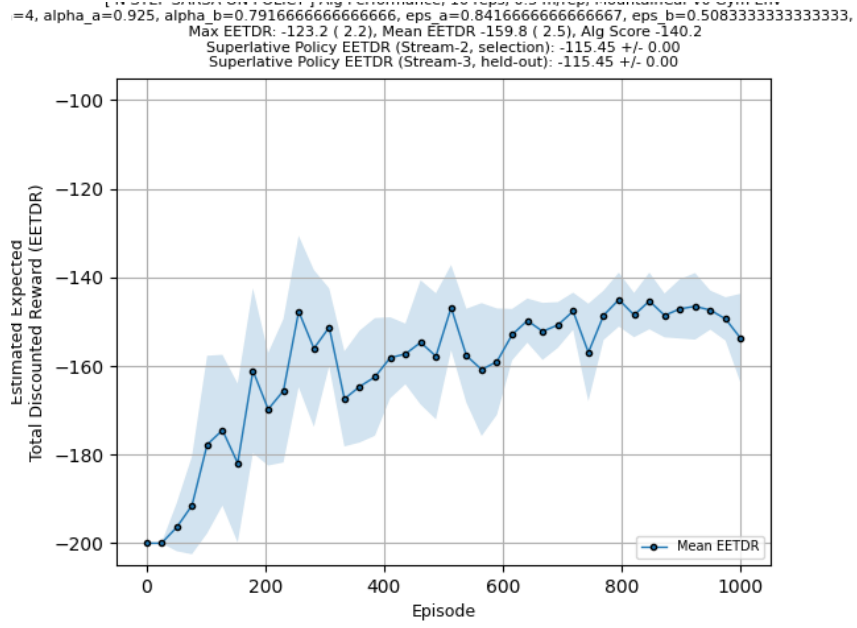


Figure 3: Learning Curve for Best n -step SARSA run

The OLS and ANOVA results are shown below.

n -step SARSA OLS and ANOVA Regression Results

(a) n -step SARSA Model Fit Summary

Metric	Value
R^2	0.879
Adj. R^2	0.776
F -statistic	8.590
Prob (F -stat)	2.20×10^{-08}
Log-Likelihood	-196.33
AIC	448.7
BIC	507.3

(b) n -step SARSA ANOVA Type II Effects Table

Variable	sum_sq	df	F	$\text{PR}(> F)$
α_a	71.717430	1.0	0.939651	0.339638
α_b	2.149772	1.0	0.028167	0.867774
ϵ_a	84.441042	1.0	1.106357	0.300754
ϵ_b	60.694407	1.0	0.795226	0.379181
qinit	55.846242	1.0	0.731704	0.398691
n_steps	288.635678	1.0	3.781740	0.060648

(c) n -step SARSA Coefficient Estimates

Variable	coef	std err	t	$P > t $	[0.025	0.975]
Intercept	-153.1315	15.298	-10.010	0.000	-184.293	-121.970
α_a	-29.3572	30.285	-0.969	0.340	-91.046	32.332
α_b	-4.4412	26.463	-0.168	0.868	-58.344	49.461
ϵ_a	-27.6490	26.286	-1.052	0.301	-81.193	25.895
ϵ_b	22.4809	25.210	0.892	0.379	-28.870	73.832
qinit	24.5221	28.668	0.855	0.399	-33.872	82.916
n_steps	49.9917	25.707	1.945	0.061	-2.372	102.355

From these results the focus shifts from ϵ_b to n-steps with the highest F-statistic and the lowest p-value. This is due to its power in balancing bias and variance as well as altering the nature of the target the agent is trying to learn. We also see an increase in R^2 values compared to the other two methods.

5 Algorithm Comparison

Table 7: Executive Summary: Model Fit and Dominant Parameter Effects Across Architectures

RL Architecture	Overall Model Fit Metrics				Primary Driver Performance Impact			
	R^2	Adj. R^2	F-Statistic	Prob (F -stat)	Factor	Coefficient	t -stat	$P > t $
On-Policy SARSA	0.767	0.647	6.415	4.09×10^{-07}	ϵ_b	80.6979	2.844	0.007**
Off-Policy Q-Learning	0.786	0.676	7.168	9.38×10^{-08}	ϵ_b	91.5923	3.374	0.002**
n-Step SARSA ($n = 6$)	0.879	0.776	8.590	2.20×10^{-08}	n.steps	49.9917	1.945	0.061 [†]

[†] Significant at the 10% level; * Significant at the 5% level; ** Significant at the 1% level.

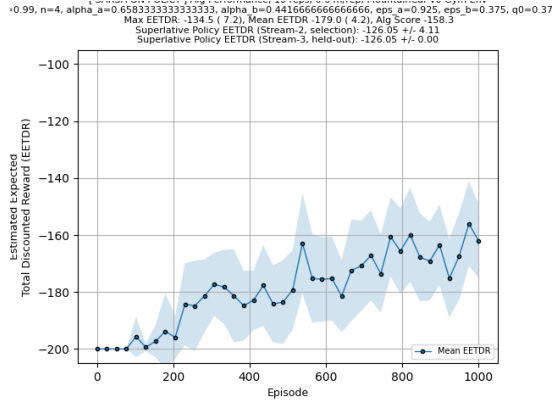
From the above table we can see that all of the algorithms have a well fit model with high R^2 and Adj. R^2 values as well as highly significant F-statistics. The dominant driver in the single step methods is ϵ_b which helps these single step methods better balance exploration and exploitation. Unless told otherwise these methods act greedy or don't with no regard to what has come before. The n-step method however is driven by the number of time steps used in the update which in combination with a good decay rate balancing exploration/exploitation balances the bias and variance in the model. This is ideal for this environment since it is a delayed reward environment where the agent needs to learn from the long term consequences of its actions. We see above that the n-step SARSA method has both the tightest variance shading, highest EETDR values, and learns the quickest. Not only that but from previous tables we see the the n-step method is also the fastest to run at 200 seconds per run. For computation time and learning performance the n-step method is the algorithm that is best for this environment. If this were a more reward dense environment the other methods may out perform the n-step method but other tests are needed to confirm this.

6 Momentum Policy Comparison

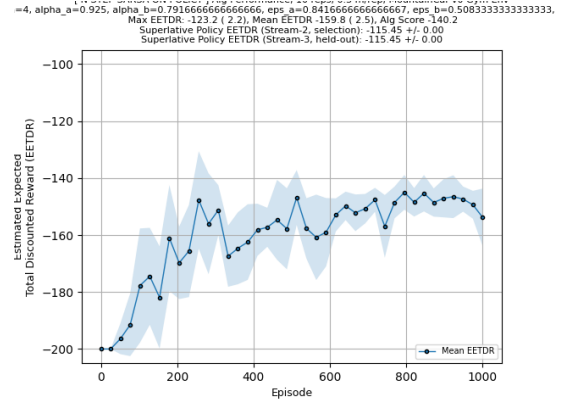
One policy to compare is a fixed policy that is defined as follows:

$$\pi(s) = \begin{cases} 0 & \text{if } s^{velocity} < 0 \\ 2 & \text{otherwise} \end{cases} \quad (1)$$

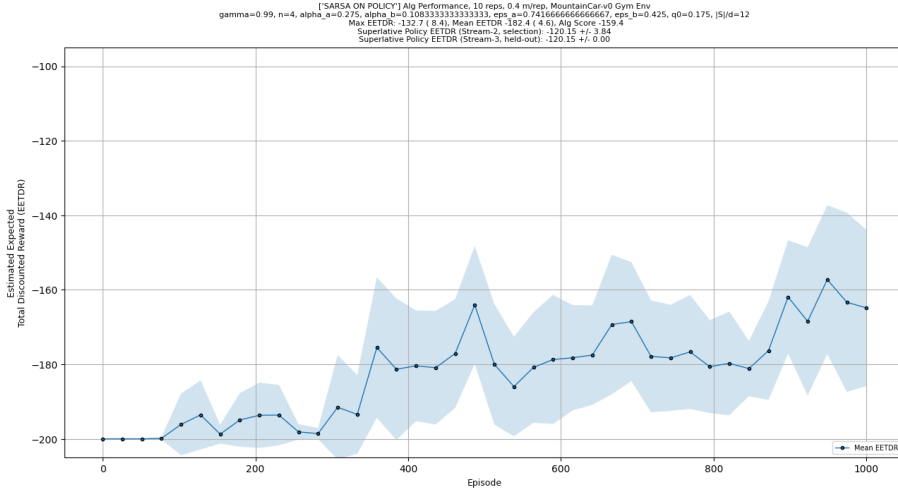
Statistically speaking this policy ties with both SARSA and Q-learning but is outperformed by the n-step SARSA method. Since it is a fixed policy it also only looks at the immediate situation and state of the car much like the SARSA and Q-learning methods. Again, the agent greatly benefits from having a



(a) Off-Policy Q-Learning (Run 27.0)



(b) Multi-Step n -Step SARSA (Run 30.0)



(c) On-Policy SARSA (Run 6.0)

Figure 4: Comparative evaluation of learning curves on the MountainCar-v0 environment. The top row contrasts standard 1-step methods, while the bottom row represents the optimized multi-step horizon architecture.

certain period to learn from and not just the immediate surroundings. The largest difference between the implementations performed above and the momentum policy can be seen in the box and whisker plot below.

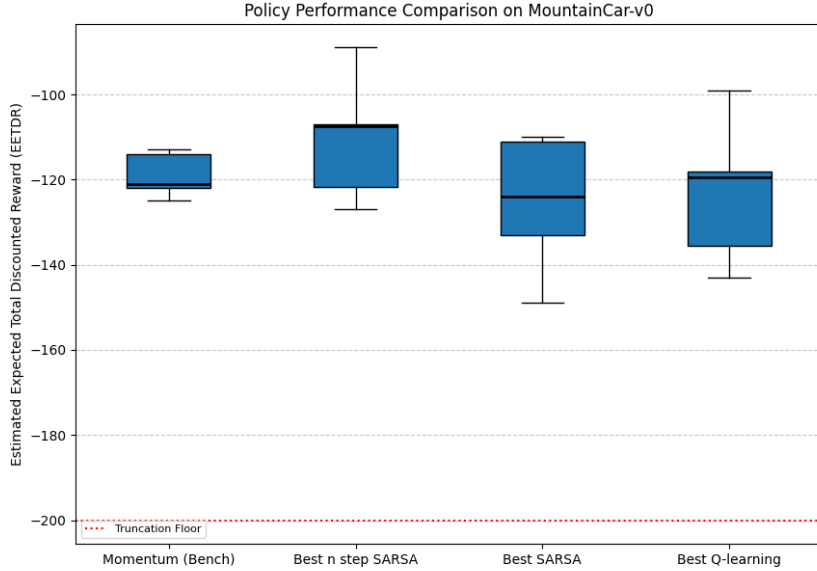


Figure 5: Box and Whisker Plot comparing the EETDR values of the Momentum Policy with the Superlative Policies from each algorithm.

Since the policy is completely fixed it has very little variance as is seen by the tight formation. The policy did not do a terrible job with an average score across the 100 episodes of -119.46 and a 95% half width of 0.74.

7 Reflection

Overall implementing these algorithms to solve the mountain car problem was not extremely difficult due to provided resources. However, ensuring that a balance of hyperparameter tuning design runs and computation time was the largest bottleneck in this project. Each LHD experiment took almost an hour apiece with 60 design runs running in parallel on 10 cores. Since this is a very sparse reward environment and very little to tell the agent how to behave when it is trying to build momentum, novel approaches such as tile coding or the implementation of neural networks may greatly improve the performance since there is more to train in those situations rather than just immediate or short-term rewards.

8 AI Usage

The use of generative AI was within class guidelines as outlined in the syllabus. It was used heavily to check syntax and coding logic during development by pasting given errors and lines of code in question into the chat. In those cases where code was directly passed to the chat the phrase, ” without editing the code...” was used. The heaviest use was in the creation of proper latex syntax for tables and figures to ensure a clean and readable document. The primary model used was Gemini 3 with some use of Copilot.

9 Appendix: Reinforcement Learning Hyperparameter Design Runs & Statistical Reports

1. On-Policy SARSA Experimental Space

Table 8: Top 10 Hyperparameter Configurations for On-Policy SARSA (Sorted by Alg Score)

Run Index	α_a	α_b	ϵ_a	ϵ_b	qinit	Sup EETDR	Sup EETDR hw	Mean Max EETDR	Mean Max EETDR ME	Time-Avg EETDR	Time-Avg EETDR ME	Secs per run	Alg Score
15.0	0.241667	0.125000	0.375000	0.358333	0.141667	-124.25	6.391664	-123.84	5.793990	-168.540875	4.702179	233.409046	-147.077616
14.0	0.591667	0.175000	0.358333	0.275000	0.941667	-114.70	7.026125	-121.72	6.493281	-173.996875	4.240568	239.248252	-148.222946
11.0	0.558333	0.341667	0.858333	0.308333	0.308333	-120.90	6.641154	-126.61	7.049661	-172.213250	3.283964	237.255877	-150.394682
41.0	0.991667	0.358333	0.175000	0.325000	0.541667	-125.95	6.997147	-131.36	8.751936	-161.449250	4.938806	222.037635	-150.622384
43.0	0.825000	0.508333	0.041667	0.758333	0.491667	-130.30	13.072783	-128.26	6.269866	-172.980625	4.664217	233.549276	-151.775856
6.0	0.275000	0.108333	0.741667	0.425000	0.175000	-127.65	5.669851	-126.75	7.230837	-174.418375	4.187460	241.776106	-151.830836
39.0	0.841667	0.541667	0.625000	0.908333	0.225000	-115.05	6.835465	-125.83	7.910474	-177.919750	4.741719	238.083015	-153.308872
21.0	0.341667	0.141667	0.125000	0.825000	0.441667	-127.10	6.088733	-136.03	9.456172	-160.759000	4.918814	221.155940	-153.562829
50.0	0.091667	0.008333	0.008333	0.341667	0.325000	-127.80	5.330998	-129.96	7.007613	-171.059625	8.509056	233.948454	-154.368040
42.0	0.441667	0.391667	0.025000	0.441667	0.608333	-115.85	7.016561	-130.89	8.335031	-173.128875	4.163171	234.302523	-154.451857

(a) SARSA Parameter Coefficient Estimates ($R^2 = 0.767$)

Variable	coef	std err	t	P > t	[0.025	0.975]
Intercept	-189.2700	14.027	-13.494	0.000	-217.642	-160.899
α_a	28.6620	33.604	0.853	0.399	-39.310	96.633
α_b	-26.5450	29.073	-0.913	0.367	-85.351	32.261
ϵ_a	24.7839	29.866	0.830	0.412	-35.626	85.194
ϵ_b	80.6979	28.378	2.844	0.007	23.298	138.098
q_{init}	23.9251	32.753	0.730	0.469	-42.324	90.174
α_a^2	-11.9534	25.464	-0.469	0.641	-63.460	39.553
$\alpha_a\alpha_b$	25.8890	20.329	1.273	0.210	-15.230	67.008
$\alpha_a\epsilon_a$	-44.1685	22.349	-1.976	0.055	-89.373	1.036
$\alpha_a\epsilon_b$	6.0655	18.713	0.324	0.748	-31.785	43.916
α_aq_{init}	7.4305	19.602	0.379	0.707	-32.219	47.080
α_b^2	-52.6465	26.771	-1.967	0.056	-106.797	1.504
$\alpha_b\epsilon_a$	7.4653	21.362	0.349	0.729	-35.743	50.673
$\alpha_b\epsilon_b$	-20.0225	19.006	-1.053	0.299	-58.466	18.421
α_bq_{init}	40.7645	20.211	2.017	0.051	-0.116	81.645
ϵ_a^2	-3.4979	23.429	-0.149	0.882	-50.888	43.892
$\epsilon_a\epsilon_b$	12.1785	21.865	0.557	0.581	-32.047	56.404
ϵ_aq_{init}	-1.1008	23.129	-0.048	0.962	-47.883	45.682
ϵ_b^2	-59.3983	23.446	-2.533	0.015	-106.822	-11.975
ϵ_bq_{init}	-23.9059	18.882	-1.266	0.213	-62.098	14.286
q_{init}^2	-40.6317	21.923	-1.853	0.071	-84.975	3.711

(b) SARSA ANOVA Type II Breakdown

Variable	sum_sq	df	F	PR(> F)
α_a	91.614265	1.0	0.727474	0.398911
α_b	104.985559	1.0	0.833650	0.366829
ϵ_a	86.721681	1.0	0.688624	0.411684
ϵ_b	1018.372770	1.0	8.086510	0.007065
q_{init}	67.197389	1.0	0.533589	0.469468
α_a^2	27.749936	1.0	0.220352	0.641386
$\alpha_a\alpha_b$	204.239872	1.0	1.621791	0.210380
$\alpha_a\epsilon_a$	491.891697	1.0	3.905924	0.055219
$\alpha_a\epsilon_b$	13.230765	1.0	0.105060	0.747571
α_aq_{init}	18.095788	1.0	0.143692	0.706694
α_b^2	487.018217	1.0	3.867226	0.056383
$\alpha_b\epsilon_a$	15.380288	1.0	0.122129	0.728617
$\alpha_b\epsilon_b$	139.762628	1.0	1.109802	0.298608
α_bq_{init}	512.310105	1.0	4.068059	0.050623
ϵ_a^2	2.806916	1.0	0.022289	0.882091
$\epsilon_a\epsilon_b$	39.069567	1.0	0.310237	0.580718
ϵ_aq_{init}	0.285276	1.0	0.002265	0.962282
ϵ_b^2	808.292719	1.0	6.418344	0.015426
ϵ_bq_{init}	201.865522	1.0	1.602937	0.212998
q_{init}^2	432.594711	1.0	3.435070	0.071402
Residual	4911.456033	39.0	NaN	NaN

2. Off-Policy Q-Learning Experimental Space

Table 10: Top 10 Hyperparameter Configurations for Off-Policy Q-Learning (Sorted by Alg Score)

Run Index	α_a	α_b	ϵ_a	ϵ_b	qinit	Sup EETDR	Sup EETDR hw	Mean Max EETDR	Mean Max EETDR ME	Time-Avg EETDR	Time-Avg EETDR ME	Sees per run	Alg Score
43.0	0.825000	0.508333	0.041667	0.758333	0.491667	-123.95	6.028940	-126.88	5.604704	-168.781000	4.182691	250.748533	-148.676299
15.0	0.241667	0.125000	0.375000	0.358333	0.141667	-127.95	6.586378	-123.90	4.504471	-174.078625	5.080434	263.673305	-148.706306
14.0	0.391667	0.175000	0.358333	0.275000	0.941667	-118.80	7.692660	-124.49	6.094680	-176.659625	4.090192	269.780487	-150.650735
53.0	0.691667	0.325000	0.575000	0.875000	0.458333	-118.60	6.044075	-126.10	6.346384	-174.840875	3.157201	258.702947	-150.667061
11.0	0.558333	0.341667	0.858333	0.308333	0.308333	-125.15	8.679411	-127.61	8.246403	-170.753250	4.204186	262.616913	-151.496816
41.0	0.991667	0.358333	0.175000	0.325000	0.541667	-127.55	5.287298	-137.33	4.676498	-160.762500	5.582127	242.912153	-151.741750
27.0	0.658333	0.441667	0.925000	0.375000	0.375000	-123.30	8.339121	-132.00	6.896231	-170.493250	2.638023	258.634193	-152.500247
36.0	0.191667	0.241667	0.225000	0.608333	0.191667	-119.40	8.021039	-130.35	7.251462	-171.768125	3.981502	256.396496	-152.860728
42.0	0.441667	0.391667	0.025000	0.441667	0.608333	-119.40	5.838394	-130.76	7.386290	-171.733250	3.640263	252.681322	-153.061179
32.0	0.075000	0.041667	0.308333	0.175000	0.475000	-118.95	7.355385	-129.34	6.707078	-175.987375	2.929779	266.639413	-153.195109

(a) Q-Learning Parameter Coefficient Estimates ($R^2 = 0.786$)

Variable	coef	std err	t	P > t	[0.025	0.975]
Intercept	-196.5146	13.419	-14.645	0.000	-223.657	-169.372
α_a	44.3574	32.149	1.380	0.176	-20.669	109.384
α_b	-17.2164	27.814	-0.619	0.540	-73.475	39.042
ϵ_a	23.0364	28.572	0.806	0.425	-34.756	80.829
ϵ_b	91.5923	27.149	3.374	0.002	36.679	146.506
q_{init}	30.3175	31.334	0.968	0.339	-33.062	93.697
α_a^2	-21.5242	24.361	-0.884	0.382	-70.800	27.751
$\alpha_a\alpha_b$	27.9053	19.448	1.435	0.159	-11.433	67.243
$\alpha_a\epsilon_a$	-44.5845	21.380	-2.085	0.044	-87.831	-1.339
$\alpha_a\epsilon_b$	2.4875	17.902	0.139	0.890	-33.723	38.698
α_aq_{init}	-1.2667	18.753	-0.068	0.946	-39.198	36.665
α_b^2	-66.3801	25.612	-2.592	0.013	-118.184	-14.576
$\alpha_b\epsilon_a$	20.8138	20.436	1.018	0.315	-20.522	62.150
$\alpha_b\epsilon_b$	-24.6954	18.183	-1.358	0.182	-61.474	12.083
α_bq_{init}	42.0229	19.335	2.173	0.036	2.913	81.132
ϵ_a^2	-8.4450	22.414	-0.377	0.708	-53.782	36.892
$\epsilon_a\epsilon_b$	8.9894	20.918	0.430	0.670	-33.320	51.299
ϵ_aq_{init}	1.0620	22.127	0.048	0.962	-43.694	45.818
ϵ_b^2	-65.6383	22.430	-2.926	0.006	-111.007	-20.269
ϵ_bq_{init}	-20.5097	18.064	-1.135	0.263	-57.047	16.028
q_{init}^2	-45.4899	20.973	-2.169	0.036	-87.912	-3.068

(b) Q-Learning ANOVA Type II Breakdown

Variable	sum_sq	df	F	PR(> F)
α_a	219.424071	1.0	1.903737	0.175520
α_b	44.162065	1.0	0.383153	0.539520
ϵ_a	74.923662	1.0	0.650042	0.424987
ϵ_b	1311.898780	1.0	11.382117	0.001687
q_{init}	107.901916	1.0	0.936164	0.339229
α_a^2	89.977193	1.0	0.780648	0.382357
$\alpha_a\alpha_b$	237.293216	1.0	2.058771	0.159303
$\alpha_a\epsilon_a$	501.202912	1.0	4.348468	0.043635
$\alpha_a\epsilon_b$	2.225252	1.0	0.019306	0.890207
α_aq_{init}	0.525871	1.0	0.004562	0.946492
α_b^2	774.250879	1.0	6.717450	0.013367
$\alpha_b\epsilon_a$	119.558384	1.0	1.037296	0.314728
$\alpha_b\epsilon_b$	212.610888	1.0	1.844626	0.182215
α_bq_{init}	544.427902	1.0	4.723491	0.035889
ϵ_a^2	16.361492	1.0	0.141953	0.708389
$\epsilon_a\epsilon_b$	21.287138	1.0	0.184689	0.669740
ϵ_aq_{init}	0.265540	1.0	0.002304	0.961962
ϵ_b^2	987.042306	1.0	8.563642	0.005694
ϵ_bq_{init}	148.583913	1.0	1.289123	0.263139
q_{init}^2	542.226002	1.0	4.704387	0.036245
Residual	4495.126115	39.0	NaN	NaN

3. Multi-Step (n -step) SARSA Experimental Space

Table 12: Top 10 Hyperparameter Configurations for n -step SARSA (Sorted by Alg Score)

Run Index	α_a	α_b	ϵ_a	ϵ_b	q_{init}	n_steps	Sup EETDR	Sup EETDR lw	Mean Max EETDR	Mean Max EETDR ME	Time-Avg EETDR	Time-Avg EETDR ME	Secs per run	Alg Score
11.0	0.558333	0.341667	0.858333	0.308333	0.308333	5.0	-111.90	0.0	-122.73	7.452758	-145.830625	5.648526	200.855438	-127.731660
20.0	0.875000	0.325000	0.141667	0.391667	0.958333	8.0	-119.55	0.0	-131.06	9.295134	-140.679750	9.015519	184.464579	-131.608108
41.0	0.991667	0.358333	0.175000	0.325000	0.541667	3.0	-118.10	0.0	-130.99	8.046902	-147.120125	5.348813	187.638711	-131.847575
27.0	0.658333	0.441667	0.925000	0.375000	0.375000	7.0	-119.00	0.0	-138.25	6.895916	-146.375375	6.672852	193.529166	-132.619691
1.0	0.425000	0.625000	0.458333	0.625000	0.158333	9.0	-116.35	0.0	-139.03	12.528568	-151.153625	7.306442	194.182282	-133.194027
31.0	0.625000	0.558333	0.158333	0.658333	0.808333	3.0	-117.55	0.0	-121.21	3.039561	-153.134875	3.660002	193.852888	-133.247951
15.0	0.241667	0.125000	0.375000	0.358333	0.141667	6.0	-118.55	0.0	-137.99	7.957033	-150.899875	4.874403	198.905594	-133.439711
30.0	0.925000	0.791667	0.841667	0.508333	0.725000	6.0	-116.95	0.0	-124.29	7.357637	-155.864875	3.670496	198.145654	-133.984148
6.0	0.275000	0.108333	0.741667	0.425000	0.175000	4.0	-116.35	0.0	-129.55	6.894186	-156.351000	4.376616	202.107249	-134.101046
36.0	0.191667	0.241667	0.225000	0.608333	0.191667	2.0	-117.20	0.0	-131.56	8.738402	-153.867750	7.268764	192.786276	-134.774605

(a) n -step Parameter Coefficient Estimates ($R^2 = 0.879$)

Variable	coef	std err	t	$P > t $	[0.025	0.975]
Intercept	-153.1315	15.298	-10.010	0.000	-184.293	-121.970
α_a	-29.3572	30.285	-0.969	0.340	-91.046	32.332
α_b	-4.4412	26.463	-0.168	0.868	-58.344	49.461
ϵ_a	-27.6490	26.286	-1.052	0.301	-81.193	25.895
ϵ_b	22.4809	25.210	0.892	0.379	-28.870	73.832
q_{init}	24.5221	28.668	0.855	0.399	-33.872	82.916
n_steps	49.9917	25.707	1.945	0.061	-2.372	102.355
α_a^2	-41.8754	22.165	-1.889	0.068	-87.025	3.274
$\alpha_a \alpha_b$	87.8747	18.134	4.846	0.000	50.937	124.812
$\alpha_a \epsilon_a$	-15.1410	20.682	-0.732	0.469	-57.269	26.987
$\alpha_a \epsilon_b$	15.6803	16.791	0.934	0.357	-18.521	49.882
$\alpha_a q_{init}$	56.9457	19.890	2.863	0.007	16.430	97.461
$\alpha_a n_steps$	26.5185	18.294	1.450	0.157	-10.746	63.783
α_b^2	-90.4788	21.630	-4.183	0.000	-134.538	-46.420
$\alpha_b \epsilon_a$	18.1634	22.982	0.790	0.435	-28.650	64.977
$\alpha_b \epsilon_b$	16.1171	15.929	1.012	0.319	-16.329	48.563
$\alpha_b q_{init}$	2.6741	17.800	0.150	0.882	-33.582	38.931
$\alpha_b n_steps$	26.8467	18.463	1.454	0.156	-10.762	64.455
ϵ_a^2	-0.7083	22.105	-0.032	0.975	-45.734	44.317
$\epsilon_a \epsilon_b$	57.3320	19.319	2.968	0.006	17.980	96.684
$\epsilon_a q_{init}$	-17.2797	21.249	-0.813	0.422	-60.562	26.002
$\epsilon_a n_steps$	12.8941	22.881	0.564	0.577	-33.713	59.501
ϵ_b^2	-70.2489	19.765	-3.554	0.001	-110.509	-29.989
$\epsilon_b q_{init}$	10.3688	15.508	0.669	0.509	-21.219	41.957
$\epsilon_b n_steps$	-7.7033	16.946	-0.455	0.652	-42.221	26.814
q_{init}^2	-46.9790	18.921	-2.483	0.018	-85.519	-8.439
$q_{init} n_steps$	-8.8474	16.210	-0.546	0.589	-41.866	24.171
n_steps^2	-54.3171	18.164	-2.990	0.005	-91.316	-17.319

(b) n -step ANOVA Type II Breakdown

Variable	sum_sq	df	F	PR(> F)
α_a	71.717430	1.0	0.939651	0.339638
α_b	2.149772	1.0	0.028167	0.867774
ϵ_a	84.441042	1.0	1.106357	0.300754
ϵ_b	60.694407	1.0	0.795226	0.379181
q_{init}	55.846242	1.0	0.731704	0.398691
n_steps	288.635678	1.0	3.781740	0.060648
α_a^2	272.410903	1.0	3.569161	0.067953
$\alpha_a \alpha_b$	1792.303811	1.0	23.482984	0.000031
$\alpha_a \epsilon_a$	40.905487	1.0	0.535949	0.469445
$\alpha_a \epsilon_b$	66.562139	1.0	0.872105	0.357367
$\alpha_a q_{init}$	625.596987	1.0	8.196648	0.007345
$\alpha_a n_steps$	160.370097	1.0	2.101189	0.156915
α_b^2	1335.459918	1.0	17.497359	0.000209
$\alpha_b \epsilon_a$	47.672283	1.0	0.624608	0.435158
$\alpha_b \epsilon_b$	78.139493	1.0	1.023793	0.319211
$\alpha_b q_{init}$	1.722695	1.0	0.022571	0.881521
$\alpha_b n_steps$	161.370031	1.0	2.114290	0.155668
ϵ_a^2	0.078366	1.0	0.001027	0.974637
$\epsilon_a \epsilon_b$	672.154739	1.0	8.806654	0.005642
$\epsilon_a q_{init}$	50.473985	1.0	0.661316	0.422105
$\epsilon_a n_steps$	24.237495	1.0	0.317563	0.577004
ϵ_b^2	964.129438	1.0	12.632142	0.001202
$\epsilon_b q_{init}$	34.121275	1.0	0.447061	0.508529
$\epsilon_b n_steps$	15.771992	1.0	0.206647	0.652477
q_{init}^2	470.532258	1.0	6.164971	0.018462
$q_{init} n_steps$	22.736600	1.0	0.297898	0.588988
n_steps^2	682.520106	1.0	8.942462	0.005324
Residual	2442.352384	32.0	NaN	NaN